Mohammed Ali Khan

+91 6305357392 | alikhanwork649@gmail.com | LinkedIn | GitHub | LeetCode

Objective

B.E. Information Technology graduate (CGPA: 8.3) with a deep focus on neural networks and deep learning. I implement architectures from first principles — understanding the math, not just the APIs. Complementary experience in web development and strong algorithmic foundations (110+ LeetCode problems). Seeking roles in **Deep Learning, AI, or Applied ML**.

Technical Skills

ML / AI : Neural Networks, CNNs, LSTMs, GRUs, Backpropagation, Gradient Descent, NumPy
Languages : Python, Java, C++, C
Web : HTML, CSS, Node.js, EJS
Blockchain : Solidity, Hardhat, ERC-20
Tools & DB : Git, GitHub, SQL

Experience

Moral Merry School — **March 2025 – July 2025**
Web Developer Intern — *Hyderabad*

- Designed and deployed a production multi-page school website with Google Maps integration, server-side dynamic rendering via EJS, and an embedded AI chatbot.
- Sole developer — handled architecture, implementation, and deployment end-to-end. Live at moralmerry.in | Certificate

Projects

Sustainable Waste Classification System | *Python, PyTorch, U-Net, Computer Vision* — **Kaggle**

- Engineered an AI-driven waste segregation system using deep learning and computer vision to enhance recycling efficiency and support environmental sustainability.

Cat vs Non-Cat Classifier | *Python, NumPy, Neural Networks* — **GitHub**

- Built a multi-layer deep binary image classifier from scratch using **only NumPy** — no ML frameworks. Manually implemented forward/backward propagation, sigmoid activations, and gradient descent.
- Demonstrates first-principles understanding of how neural networks learn, not just API-level usage.

StableX – Crypto Token Stabilizer | *Solidity, Hardhat, Python* — **GitHub** | **Video**

- Engineered a decentralized ERC-20 token on a local blockchain with a Python bot that monitors price volatility and autonomously executes stabilization logic.

Technical Writing

Why LSTM Works Better Than Vanilla RNN | *Medium* — **Read Article**

- Technical deep-dive on the vanishing gradient problem in RNNs and how LSTM forget, input, and output gates enable long-term dependency learning — written for practitioners and students alike.

Certifications

Deep Learning Specialization – DeepLearning.AI (Coursera) | Verify

- 5-course program covering neural networks, CNNs, sequence models (LSTM/GRU), optimization, and ML project structuring.

TCS ION NQT – IT (Valid till 2027) | Foundation: **80.31%** Verify Psychometric: **90%** Verify

Education

B.E. Information Technology — Muffakham Jah College of Engineering & Technology — **2022 – 2026**
*CGPA: **8.3 / 10*** | *Transcripts* — *Hyderabad, India*